AIST3120 Introduction to NLP
Project Report

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Abstract

This project investigates techniques for Named Entity Recognition (NER) on the CoNLL-2003 benchmark, combining RoBERTa-large with a Conditional Random Field (CRF) layer and Focal Loss to address class imbalance and tokenization challenges. We propose a data augmentation strategy to mitigate inconsistencies in RoBERTa's whitespace-sensitive tokenization, particularly for entities at sentence boundaries. Experiments demonstrate that Focal Loss outperforms Cross-Entropy and Dice Loss, especially for minority classes, while the CRF layer improves label coherence. Our best model achieves strong performance, though limitations in token-level predictions suggest future work on span-based approaches.

1 Introduction

The CoNLL-2003 dataset (Tjong Kim Sang and De Meulder, 2003) is a widely used benchmark for evaluating named entity recognition (NER) systems. Released as part of the CoNLL-2003 Shared Task, it focuses on language-independent NER and includes annotated data for English and German. The dataset defines four types of named entities: persons (PER), organizations (ORG), locations (LOC), and miscellaneous entities (MISC) that do not fall into the other categories. The **MISC** category, introduced in this dataset, captures named entities that do not fit into the other three classes, such as nationalities, religious groups, events, or product names, providing a broader coverage of entity types in real-world texts.

One of the most advanced models for named entity recognition is the Automated Concatenation of Embeddings for Structured Prediction(ACE) (Wang et al., 2021), which achieved a state-of-the-art F-1 score of 94.6 on the CoNLL-2003 English test set as of 2021. This performance was attained by leveraging ACE embeddings combined with document-level context, allowing the model to capture richer semantic and contextual information for more accurate entity recognition.

During our experiments, the primary challenge was accurately identifying the **MISC** and **ORG** entity types. While achieving a high F_1-score was relatively straightforward for other categories, the performance for **MISC** and **ORG** remained consistently lower regardless of the approaches used. The **MISC** label is often applied to ambiguous words such as "Apple", which can refer to an **ORG** (company) or a **MISC** (fruit). Accurately identifying such words is crucial, as they may mislead the model into incorrect classifications. Therefore, instead of masking these words as non-entity tokens, we guess that this is a reason of CoNLL-2003 propose introducing a new class to specifically classify such ambiguous terms.

2 Related work

2.1 State-of-the-Art: Automated Concatenation of Embeddings

Recent advances in Named Entity Recognition have explored automated methods for combining different embedding types to improve performance. A novel approach is proposed for structured prediction tasks that automatically concatenates multiple embeddings without manual feature engineering (Wang et al., 2021). Their method dynamically combines contextual embeddings (like BERT) with traditional static embeddings (such as GloVe) through learned weighting mechanisms. This approach demonstrated significant improvements on several NER benchmarks, particularly in handling rare entities and domain adaptation scenarios. The automated concatenation framework addresses key limitations of single-embedding models by preserving both the contextual understanding from modern language models and the lexical information from traditional embeddings.

2.2 BERT: Bidirectional Encoder Representation from Transformers

The introduction of BERT (Devlin et al., 2019) revolutionized the field of natural language processing and established new state-of-the-art performance for NER tasks. BERT's key innovation lies in its deep bidirectional Transformer architecture, which is pre-trained using masked language modeling and next sentence prediction objectives. Unlike previous approaches that processed text sequentially, BERT learns contextual representations by attending to all words in a sentence simultaneously. This bidirectional understanding proved particularly valuable for NER, where entity recognition often depends on global context. The original BERT model achieved remarkable results on standard benchmarks like CoNLL-2003, demonstrating the effectiveness of transfer learning for sequence labeling tasks. However, BERT's computational requirements and fixed-length attention span presented challenges for certain NER applications.

2.3 RoBERTa: A Robustly Optimized BERT

Building upon BERT's foundation, RoBERTa was introduced, which refined the pre-training approach through several key modifications (Liu et al., 2019). RoBERTa removed the next sentence prediction objective, implemented dynamic masking patterns during training, and used significantly larger batches and more training data. These optimizations resulted in improved performance across various NLP tasks, including NER. For entity recognition specifically, RoBERTa demonstrated better handling of ambiguous entity boundaries and improved consistency in labeling similar entities across different contexts. The model's robust optimization strategy made it particularly effective for NER in noisy or domain-specific text, though it retained many of BERT's architectural constraints. RoBERTa's success confirmed the importance of careful pre-training procedures in developing high-performing language models for sequence labeling tasks.

3 Methods

3.1 Architecture

In this project, we employs a RoBERTa encoder with a Conditional Random Field (CRF) layer (Sutton and McCallum, 2010) for structured prediction. For the encoder component, we experimented with both RoBERTa-base (125M parameters) and RoBERTa-large (561M parameters) to compare their performance in capturing contextual token representations. Our final decision is to use RoBERTa-large. The model utilizes Focal Loss as its optimization objective to address class imbalance, particularly for rare entity categories. We also add a CRF layer to model label dependencies and enforce valid transition constraints between entity tags during sequence prediction. For the model output, we add a final MLP layer with softmax activation produces the probability distribution over entity classes.

3.1.1 Baseline

We would adapt RoBERTa as our baseline architecture. We choose RoBERTa since it has demonstrated strong performance in NER tasks (Shen et al., 2023) due to its robust pre-training on a large corpus, improved masked language modeling objective, and dynamic token masking. Additionally, RoBERTa's ability to capture deep contextual representations makes it well-suited for identifying named entities in the CoNLL-2003 dataset. Its superior performance over BERT and other variants in benchmark evaluations further justifies our selection.

3.2 Loss function

We have tried different loss functions including Dice Loss, Cross-Entropy Loss, and Focal Loss (More experiment details in section 4.3). As a result, the Focal Loss can address both the focus on hard positives and class imbalance simultaneously. Focal Loss is a variation of the cross-entropy loss, defined as follows:

$$\mathcal{L}_{\text{Focal}} = -\alpha_t (1 - p_t)^\gamma \log(p_t)$$

3.2.1 Conditional Random Field

Conditional Random Fields (CRF) is probabilistic graphical models widely used in structured prediction tasks, particularly in natural language processing (NLP) and computer vision. The target of it is to model the conditional probability of a sequence of label given observed input data. The mathematical formula of CRF can be express as:

$$P(y|x) = \frac{1}{Z(x)} \exp\left(\sum_i [t(y_{i-1}, y_i) + s(y_i, x_i)]\right)$$

where $t(y_{i-1}, y_i)$ is the transition score for label y_i and y_{i-1}, $s(y_i, x_i)$ is the emission score for label y_i at the position i (Sutton and McCallum, 2010). With transition score and emission score, CRF can both capture the dependencies between neighboring labels and the relationships of labels and input sequence, which is suitable in name entity recognition tasks.

By combining RoBERTa with a CRF layer, we enhanced the model's ability to learn label dependencies and enforce valid prediction sequences while maintaining RoBERTa's strong contextual representations.

3.3 Adopting Original Tokenization Behavior of RoBERTa

English grammar requires the first word of a sentence to be capitalized, regardless of whether it represents a named entity. This convention poses a challenge for Named Entity Recognition (NER) systems, which must distinguish between:

- Capitalization due to grammatical rules
- Capitalization indicating named entities (e.g., **Person**, **Organization**, **Location**)

RoBERTa's case-sensitive tokenization provides an effective solution by preserving capitalization patterns, enabling the model to learn these critical distinctions. The tokenizer produces different outputs for capitalized versus lowercase variants of the same word, as demonstrated below:

Table 1: Tokenization Examples with RoBERTa (Including Input IDs)

Input	**Tokenized Output**	**Input IDs**
"Peter, how are you?"	['Peter', ',', 'Ġhow', 'Ġare', 'Ġyou', '?']	[0, 22611, 6, 141, 32, 47, 116, 2]
"peter, how are you?"	['p', 'eter', ',', 'Ġhow', 'Ġare', 'Ġyou', '?']	[0, 642, 5906, 6, 141, 32, 47, 116, 2]
" Peter, how are you?"	['ĠPeter', ',', 'Ġhow', 'Ġare', 'Ġyou', '?']	[0, 2155, 6, 141, 32, 47, 116, 2]
" peter, how are you?"	['Ġp', 'eter', ',', 'Ġhow', 'Ġare', 'Ġyou', '?']	[0, 181, 5906, 6, 141, 32, 47, 116, 2]

The input IDs of different cases, such as uppercase and lowercase letters, play a crucial role in the model's processing pipeline, as they provide different numerical representations that the transformer architecture processes. Each unique tokenization pattern results in distinct input ID sequences, allowing the model to maintain awareness of capitalization differences at the most fundamental level.

In RoBERTa's tokenization, which is derived from the GPT-2 tokenizer and uses byte-level Byte-Pair Encoding, the character Ġ represents a "SPACE." It is used to identify the boundaries between tokens. Additionally, we believe that it can help identify capitalized words, as these often represent entities. When a capital letter appears following a Ġ, it typically indicates that this word is likely an entity type.

Therefore, this tokenization behavior is particularly valuable for NER tasks, as it helps the model differentiate between:

- True named entities (consistently capitalized)
- Words capitalized solely due to sentence position

This method allows the model to identify entities by learning the behavior that a capitalized word, when prefixed with the 'Ġ' character, must be an entity.

3.4 A Data Augmentation Strategy for RoBERTa's Tokenization Behavior

However, there is a disadvantage associated with the tokenization behavior of RoBERTa. Specifically, the first word in an input sequence is tokenized **without** the leading marker `Ġ`, which typically indicates a whitespace prefix in RoBERTa's byte-level BPE tokenizer. This can cause inconsistencies in how entities are represented within the model.

For instance, the phrase *"The Chinese University of Hong Kong"* may be tokenized differently depending on its position in the input. When it appears at the beginning of a sentence, the first token lacks the `Ġ` prefix (e.g., `''The''` vs. `''ĠThe''`), leading to two distinct representations for the same entity. This variation can hinder the model's ability to consistently recognize and learn entity patterns.

To mitigate this issue, we propose augmenting the dataset by introducing duplicate samples for cases where an entity appears at the beginning of a sentence. In these duplicates, we prepend a space to the input, ensuring that the first token is also prefixed with `Ġ`. For example, the original input:

```
The Chinese University of Hong Kong is the best
```

would be duplicated and modified to:

```
ĠThe Chinese University of Hong Kong is the best (which meaning add a space in
the data.)
```

This augmentation ensures that the model learns entity representations both with and without `Ġ` prefixes, improving robustness and consistency in entity recognition. Therefore, we ensure that all data starting with an entity has a copy with a `Ġ` prefix, guaranteeing that the model learns the entity in both versions. As a result of this data augmentation strategy, the size of the training dataset increases from **12,705** to **19,928** samples.

3.5 Alternative Method: Adopting All-prefix Token Tokenization

We also experimented with encoding all words using the space-prefixed token format (i.e., with the `Ġ` prefix). While this approach may make it more difficult for the model to identify tokens based on word capitalization, it offers a useful trade-off. By ensuring that **all tokens** have a `Ġ` prefix, the total number of distinct tokens the model needs to handle may be slightly reduced, and would not be confusing on between sub-word token, even if we configure the model to avoid predicting sub-word tokens.

However, this could weaken the model's ability to detect sentence boundaries or leverage positional cues, such as capitalized proper nouns at the start of a sentence, as the `Ġ` prefix usually indicates a space before the word. However, if changing to separate the first word and the sub-word token, the all tokenization strategy can be difficult.

Despite these challenges, we adopt this method of tokenizing **every word** with a `Ġ` prefix.

In this method, each **word** (not sub-word token) is prefixed with `Ġ`, including the first word in a sentence. For example:

`ĠTheĠChineseĠUniversityĠofĠHongĠKongĠisĠtheĠbest` *(For Materialization, we use Ġ for space. The first of the word will add a Ġ prefix.)*

4 Experiments

4.1 Data

This study employed two datasets to train and evaluate the Named Entity Recognition (NER) model: the CoNLL-2003 dataset and the WikiANN dataset.

CoNLL-2003 Dataset: The CoNLL-2003 dataset (Tjong Kim Sang and De Meulder, 2003) is a widely acknowledged benchmark for NER tasks. It contains annotated entities across four categories: *PER* (Person), *LOC* (Location), *ORG* (Organization), and *MISC* (Miscellaneous). This dataset was chosen for its high-quality annotations and relevance in evaluating NER systems in English.

The metric used to evaluate this dataset is the F1 score.

WikiANN Dataset: To enhance model robustness and generalization, the WikiANN dataset (Pan et al., 2017) was incorporated during pretraining. WikiANN provides multilingual and domain-diverse annotations, which complement the smaller size and narrower domain of CoNLL-2003. This inclusion enabled the model to learn more generalized representations before focusing on domain-specific fine-tuning.

After using the WikiANN dataset, the F-1 score increased from 0.9215 to 0.9280. This improvement was achieved by using a combination of RoBERTa with a Conditional Random Field (CRF) layer and focal loss.

Preprocessing Steps: The datasets were tokenized using a BERT-based tokenizer, with sub-word labels aligned to their respective original words. Special tokens such as `[CLS]` and sub-words were assigned a label of `-100` to be ignored during training. To ensure uniform input size, sequences were padded or truncated to a maximum length of 128 tokens.

Hyperparameter: The model will pre-train for `10 epochs` on WikiANN and train on Conll for `20 epochs`, optimized using the AdamW optimizer with default parameters ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1\text{e}{-8}$) (Loshchilov and Hutter, 2019). The optimization uses a learning rate of $5\text{e}{-5}$ coupled with L2 weight decay (0.01) to prevent overfitting. The learning rate follows a cosine scheduler with restarts, which dynamically adjusts AdamW's base learning rate to help escape local optima, while 500 warmup steps ensure stable gradient updates during early training when the optimizer's momentum estimates are being initialized.

The alpha of focal loss have been set to 3.

4.2 Metric

In this project, we adopted F-1 score as our evaluation metric. The mathematical formula of F-1 score is given by

$$F_1 = \frac{2 \times precision \times recall}{precision + recall}$$

which is the harmonic mean of precision and recall. Precision is the proportion of system-identified named entities that are correct. Recall is the proportion of actual named entities in the dataset that the system correctly identifies. We consider a named entity is correct only if it exactly matches the corresponding entity in the data.

In our results, we will present three F1 scores: Micro Average, Macro Average, and Weighted Average. Unless specifically mentioned, the term "F1 score" refers to the Micro Average F1 score.

Micro Average F1 Score: is calculated by aggregating the contributions of all classes to compute the average metric. It treats each instance equally, making it particularly useful when we have imbalanced class distributions.

Macro Average F1 Score: The Macro Average F1 score is computed by calculating the F1 score for each class independently and then taking the average of these scores. This method treats all classes equally, regardless of their frequency, which can provide insight into the performance across all classes.

Weighted Average F1 Score: is similar to the Macro Average, but it accounts for the number of instances of each class. Each class's F1 score is weighted by the number of true instances, providing a more balanced view when class sizes vary significantly.

4.3 Loss function

In our experiments, we explored multiple loss functions, including Cross-Entropy Loss, Focal Loss, and Dice Loss, each tailored for different challenges in classification and segmentation tasks.

Cross-Entropy Loss: This loss is commonly used for classification tasks and quantifies the difference between the predicted probability distribution and the ground truth. For binary classification, it is defined as:

$$\mathcal{L}_{\mathrm{CE}} = -\left[y \log(p) + (1-y)\log(1-p)\right]$$

where $y \in \{0,1\}$ is the ground truth label and p is the predicted probability for the positive class.

We found that using Cross-entropy loss, when classes are imbalanced, may become biased towards the majority class (`PER`, `ORG`, `LOC`), leading to poor performance on minority classes (`MISC`). (Table 2)

Focal Loss: Originally proposed by Facebook AI for one-stage object detection (Lin et al., 2018), Focal Loss is particularly effective in scenarios with extreme foreground-background imbalance. The focal loss is designed to address class imbalance by **down-weighting easy examples** and **focusing on hard** ones, Focal Loss modifies the cross-entropy loss as follows:

$$\mathcal{L}_{\mathrm{Focal}} = -\alpha_t (1-p_t)^{\gamma} \log(p_t)$$

where:

- p_t is the model's estimated probability for the true class,
- α_t is a balancing factor for class t, addressing class imbalance (`MISC`),
- γ (the focusing parameter) controls how much easy examples are down-weighted. This factor reduces the loss contribution from well-classified examples (e.g., high p_t for `PER`), forcing the model to focus on harder cases (`MISC` and `ORG`).

In our experiments, we observe that the difficulty of identifying each class varies significantly. For instance, the **PER** (Person) class is very easy to predict—by the first epoch, the F-1 score already exceeds **90%**. The **focusing parameter** γ helps reduce the loss contribution from well-classified classes, allowing the model to prioritize harder examples. The value of the focusing parameter decreases the contribution of the well-classification loss. We have tried different values for the **focusing parameter**, and the best value is 3.

Additionally, there is a notable **class imbalance** in the dataset. For example, the **MISC** (Miscellaneous) class contains only half as many samples as other classes, both in the training and test sets. To address this, the α **factor** adjusts the loss function to increase the contribution of underrepresented classes, improving their recognition.

From our experiments, it demonstrate that focal loss has been shown to improve the F-1 score for minority classes. Specifically, the `MISC` achieved a **2% increase** (Table 2) in F-1 score compared to Cross-entropy loss, highlighting its effectiveness in handling class imbalance cases.

Dice Loss: Dice Loss is particularly effective for segmentation tasks, especially when dealing with imbalanced foreground-background ratios. It is derived from the Dice Coefficient and is defined as:

$$\mathcal{L}_{\mathrm{Dice}} = 1 - \frac{2\sum_i p_i y_i + \epsilon}{\sum_i p_i + \sum_i y_i + \epsilon}$$

where p_i and y_i are the predicted and ground truth values for pixel i, respectively, and ϵ is a small constant to avoid division by zero.

The Dice Loss can also be expressed in terms of the Intersection over Union (IoU) and the F-1 score, which provides an alternative perspective on its formulation:

$$\mathcal{L}_{\mathrm{Dice}} = 1 - 2\mathrm{IoU} = 1 - \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} = 1 - \mathrm{F1}$$

where TP, FP, and FN represent the true positives, false positives, and false negatives, respectively.

Despite its theoretical advantages for imbalanced data (e.g., improved focus on rare classes like `MISC`), our experiments revealed that Dice Loss underperformed Cross-Entropy in the RoBERTa+CRF framework. This difference likely comes down to two main factors: First, Dice Loss's gradient dynamics are sensitive to small prediction fluctuations, especially for low-probability tokens. This instability can disrupt RoBERTa's fine-tuning process, leading to suboptimal token representations. The CRF layer's transition constraints may further amplify this effect, as Dice Loss lacks probabilistic calibration. Second, while Dice Loss excels in segmentation, NER requires precise discrete token labels. Cross-Entropy's log-likelihood objective aligns better with the CRF's global likelihood maximization, whereas Dice Loss's non-linear normalization may conflict with the CRF's marginal probability estimates.

4.4 Results

Our best result across all methods is achieved using $\textbf{RoBERTa}_{\textbf{large}}$ combined with a Conditional Random Field (CRF) and employing focal loss for training.

In table 4, using the CRF layer improve about 1 score up compare to only using only $\textbf{RoBERTa}_{\textbf{base}}$.

Moreover, in tokenization, if we use `all-prefix tokenization` on word, data augmentation cannot be applied because we only have prefix-based tokens. When comparing the two methods, although data augmentation ensures that the model learns different representations of tokens, `all-prefix tokenization` outperforms `data augmentation + original RoBERTa tokenization` by approximately `0.5 F1-score`. This improvement occurs because the model deals with lower token complexity, making sub-word tokens and the first word more straightforward to learn.

We observe that focal loss is effective. The F1 score can reach 92.3 with $\textbf{RoBERTa}_{\textbf{base}}$ and 93.0 with $\textbf{RoBERTa}_{\textbf{large}}$. Notably, we achieve the highest F1 score in the MISC class. Although cross-entropy performs nearly as well in other entity classes, focal loss demonstrates its ability to handle minor classes, achieving an F1 score of 83 in MISC, while cross-entropy only achieves 81. Therefore, the F1 score based on the Macro Average will show a more significant difference. The Macro Average benefits from a score of 91 due to a more even distribution of average class F1 scores compared to cross-entropy.

We also find that the larger model of RoBERTa achieves higher F1 scores (see Table 4) under the same settings. The F1 score with $\textbf{RoBERTa}_{\textbf{large}}$, combined with CRF and focal loss, reaches 93, indicating that a more complex encoder model can improve performance.

The results indicate that Dice loss provides a better balance between precision and recall, as metrics are based on the F1-score. Although Dice loss results do not achieve the highest scores, detailed results in Table 3 show that Dice loss effectively balances precision and recall, offering valuable information.

Table 2: Detailed Classification Report Using Cross-Entropy Loss (left) and Focal Loss (Right) in $\textbf{RoBERTa}_{\textbf{base}}$

Label	Precision	Recall	F1 Score
LOC	0.9436	0.9340	0.9388
MISC	0.7863	0.8333	0.8091
ORG	0.8916	0.9314	0.9111
PER	0.9677	0.9641	0.9659
Micro Avg	0.9143	0.9293	0.9217
Macro Avg	0.8973	0.9157	0.9062
Weighted Avg	0.9157	0.9293	0.9223

Label	Precision	Recall	F1 Score
LOC	0.9408	0.9346	0.9377
MISC	0.8092	0.8519	0.8300
ORG	0.8864	0.9211	0.9035
PER	0.9732	0.9672	0.9702
Micro Avg	0.9137	0.9304	0.9230
Macro Avg	0.9024	0.9187	0.9103
Weighted Avg	0.9177	0.9297	0.9235

Table 3: Detailed Classification Report Using Dice Loss

Label	**Precision**	**Recall**	**F1-Score**
LOC	0.9202	0.9346	0.9273
MISC	0.8052	0.8006	0.8029
ORG	0.8844	0.8844	0.8844
PER	0.9671	0.9629	0.9650
Micro Avg	0.9088	0.9112	0.9100
Macro Avg	0.8942	0.8956	0.8949
Weighted Avg	0.9088	0.9112	0.9100

Model	**Loss**	**Data Aug + Default Tokenization**	**All Prefix Tokenization**	**F1 Score**
RoBERTa	Cross-Entropy		✓	0.911
RoBERTa	Dice		✓	0.917
RoBERTa + CRF	Cross-Entropy		✓	0.918
RoBERTa + CRF	Cross-Entropy		✓	0.922
RoBERTa + CRF	Focal		✓	0.923
$\text{RoBERTa}_{\text{Large}}$ + CRF	Dice		✓	0.910
$\text{RoBERTa}_{\text{Large}}$ + CRF	Focal	✓		0.925
$\text{RoBERTa}_{\text{Large}}$ + CRF	Focal		✓	0.930

Table 4: Model comparison with different configurations and F-1 scores

5 Conclusion

This project successfully advanced Named Entity Recognition (NER) performance on the CoNLL-2003 dataset by integrating RoBERTa-large with a Conditional Random Field (CRF) layer and Focal Loss. Our approach addressed key challenges in NER, including class imbalance and tokenization inconsistencies, achieving a F1-score of 93.0. The CRF layer enhanced label coherence, while Focal Loss significantly improved recognition of minority classes like MISC. Additionally, our all-prefix tokenization method further boosting model robustness.

6 Limitation

The limitation of our approach is the reliance on token-level predictions rather than span-based modeling, which restricts the model's ability to consistently improve F-1 scores across all entity classes simultaneously. While techniques like CRF and Focal Loss helped optimize individual class performance, the inherent trade-offs between precision and recall for different entity types, especially for ambiguous or overlapping spans is remain unresolved. A span-based framework could better capture entity boundaries and dependencies, potentially offering more balanced gains across all class categories. Future work should explore hybrid architectures to address this constraint.

7 Contribution

Wong Wai Chung, Tsang Cheuk Hang focus on model training, Tsang Wai Choi Richard focus on report writing.

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